Algebra 1 — Simplifying expressions · Brackets · Solving equations

No calculator. Marks are shown in brackets. ★ = harder question (grades 7–9). Check every solution by substituting it back into the equation.

A · SIMPLIFY AS MUCH AS POSSIBLE

1 $7pq - 2qp$ [1]

2 $3a^2b - ab^2 + 2ba^2 + 4b^2a$ [1]

3 $3c^3 \times 4c^2$ [1]

4 $2y \times (3y)^2$ [1]

5 $7m - (3n - 2m) + n$ [2]

B · SOLVE FOR x

6 $4 = \dfrac{x}{9}$ [2]

7 $4 = \dfrac{36}{x}$ [2]

8 $6(3 - 2x) = 6$ [2]

9 $5x + 7 = 31 - 3x$ [2]

10 $3(x - 1) - (x + 4) = 5$ [2]

C · PROBLEMS LEADING TO EQUATIONS

11 The sum of three consecutive even numbers is 150. What are the numbers? [3]

12 Three angles meet on a straight line. Their sizes are $(3x + 10)°$, $2x°$ and $(4x - 10)°$. Find x and the size of each angle. [3]

13 An isosceles triangle has two equal sides of length $(2x + 5)$ cm and $(4x - 7)$ cm, and a base of $3x$ cm. Find x and the perimeter of the triangle. [3]

14 The length of a rectangle is 3 cm more than its width. Its perimeter is 42 cm. Find the area of the rectangle. [3]

15 Ana is 4 years older than Ben. In 5 years' time the sum of their ages will be 40. How old is each of them now? [3]

Total: 31 marks

Check: like terms only · a minus outside a bracket changes every sign inside · "do the same to both sides" · substitute your answer back.